DIC20
Combinational and Sequential Timing

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Rebuild Lab 20 from PDF pages 81–84, distinguishing combinational propagation/transition times from sequential setup/hold, with successful and failing sweep points.

2 Specification Mapping

S denotes simulated checks; D denotes analytical discussion.

Requirement and PDF pages	Deck / check	Status
Four inverters, 10fF and edge/delay measurements (81,84)	D01/C0	S
Both data-polarity setup times (82–83)	D02; 62 cases	S
Both hold times including zero (83)	D03; 102 cases	S
Narrow-pulse propagation (84)	D01/C1–5	S
Minimum pulse/release definitions (84)	Discussion	D

3 Circuit and Design Derivation

The four inverters use $W_p/W_n/L = 3.6/1.8/0.25\,\mu$m and a final 10 fF load. The static DFF follows Fig.20.2(b): four transmission gates and four inverters form master/slave latches, with a 10 fF output load. The master is transparent at low clock and slave at high clock, giving positive-edge capture. Ideal complementary clocks isolate intrinsic timing without clock-inverter skew.

$$t_{PLH/PHL} = t_{out,50\%} - t_{in,50\%}, \quad t_r = t_{90\%} - t_{10\%}, \quad t_f = t_{10\%} - t_{90\%}.$$

Setup offset is how early the data midpoint precedes the clock midpoint; hold offset is how late it follows. A negative hold value only describes this test, not unrestricted early data changes for arbitrary clock waveforms.

4 Simulation Setup

The model is supplied separately on the user's computer. These are 1 V, 27 °C teaching conditions; the handout's old-process 2.5 V numbers are not pass thresholds. Actual PrimeSim HSPICE S-2021.09 runs use `accurate`, `runlvl=5` and `.print/.measure` data. Every `.alter` resets baseline parameters and temperature. All cases completed without errors or failed measurements; model and convergence notices may remain. Public netlists retain only `* MODEL_SETUP_REQUIRED`, with no model path. Full data are under the lab's `results/`; large files are split by deck and indexed by `simulation_data.json`.

Three decks contain 170 cases. The chain uses 5 ps input edges, widths of 1 ns and 100/50/20/10/5 ps, and at most 0.5 ps steps. DFF data/clock edges are 20 ps; the clock midpoint is 1.01 ns. Both data polarities sweep setup from -100 to 200 ps and hold from -300 to 200 ps, in 10 ps steps. DFF simulation lasts 3.5 ns with at most 1 ps steps. Capture at 1.8 ns and retention at 3.0 ns must both satisfy logic0 $\leq$ 0.1 V or logic1 $\geq$ 0.9 V.

5 Results and Specification Checks

Width (ps)	$V_{out,pk}$ (V)	t_{PLH} (ps)	t_{PHL} (ps)	t_r (ps)	t_f (ps)
1000	1.1171	269.43	310.50	87.92	44.24
100	0.7882	262.33	233.49	–	–
50	0.0089	–	–	–	–
20	0.0046	–	–	–	–
10	0.0046	–	–	–	–
5	0.0046	–	–	–	–

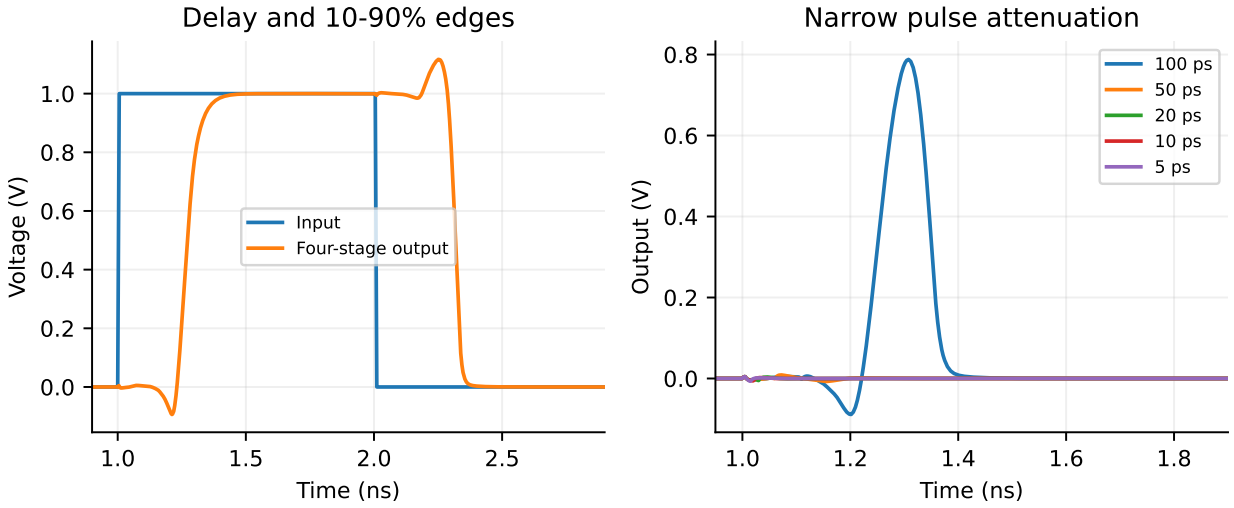


Figure 1: Four-stage timing and narrow-pulse attenuation. Actual overshoot/undershoot is retained.

Check	Captured bit	Failing offset (ps)	Passing offset (ps)
setup	0	140	150
setup	1	110	120
hold	0	-120	-110
hold	1	-150	-140

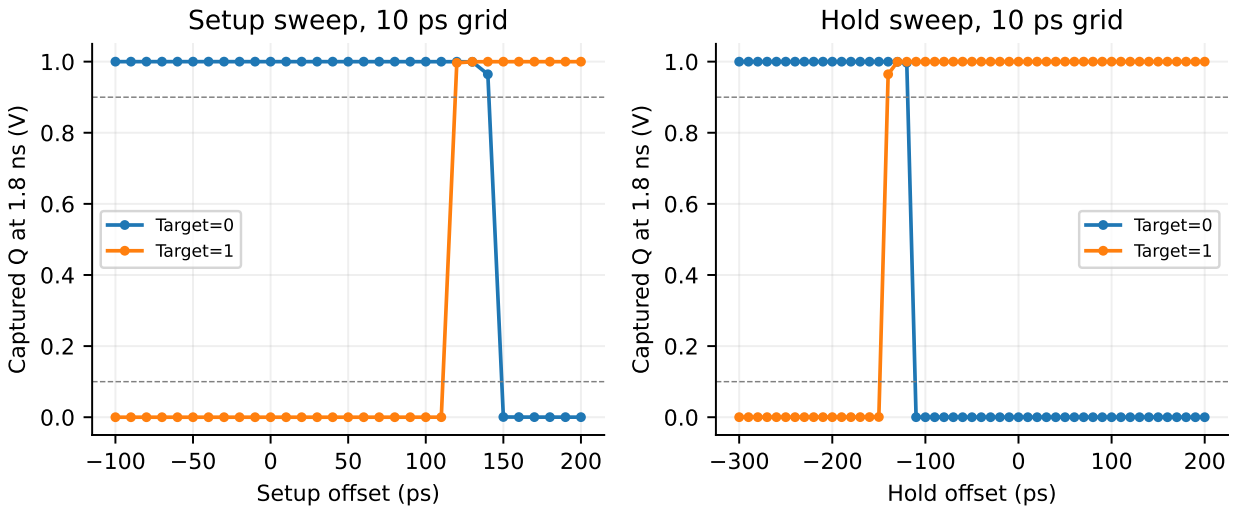


Figure 2: Captured voltage versus offset for both polarities. The pass table also checks retention at 3.0 ns.

The worst setup passing point is 150 ps, preceded by failure at 140 ps; the other polarity gives 120/110 ps. The most conservative hold passing point is -110 ps, preceded by failure at -120 ps; the other polarity gives -140/-150 ps. Both polarities pass at zero hold. These are 10 ps-grid boundaries under the declared criterion, not exact continuous-time or clock-to-Q-degradation library limits.

A 100 ps pulse peaks at 0.788 V but misses the 0.9 V valid-high criterion. Pulses of 50 ps or less produce no 50%-crossing output pulse. A dash denotes a missing crossing, not zero delay.

6 Discussion

Finite bandwidth and inertia prevent sufficiently narrow pulses from charging internal nodes to their switching thresholds before the stimulus reverses. Pulses shrink or disappear; a partial voltage excursion near the threshold is not a clean digital pulse. The supplied model with fast ideal sources produces rail excursions; they are retained without claiming reliability ratings.

Minimum pulse width is the high/low duration required to complete a specified clock or set/reset operation. Release/recovery time is the interval from asynchronous-control deassertion to an active clock edge. Fig.20.2(b) has no asynchronous set/reset pin, so these definitions are explained without inventing pin measurements. Setup and hold interact; separate sweeps with other conditions fixed are not a complete two-dimensional timing surface, PVT analysis or metastability MTBF.

7 Conclusion

Combinational timing, both setup/hold polarities and the narrow-pulse question have actual results. Worst passing setup/hold points of 150/-110 ps apply to the specified model, clock, load and criterion.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. The MOS primitives and full dff hierarchy in `02_setup_sweep.sp` and `03_hold_sweep.sp`.

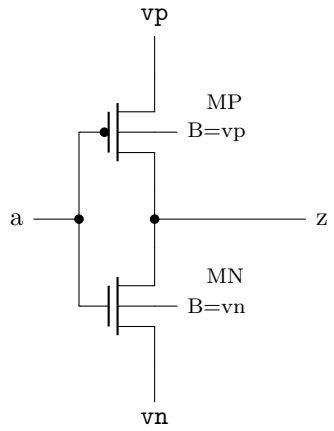


Figure 3: Exact inv cell; k=1 in the flip-flop.

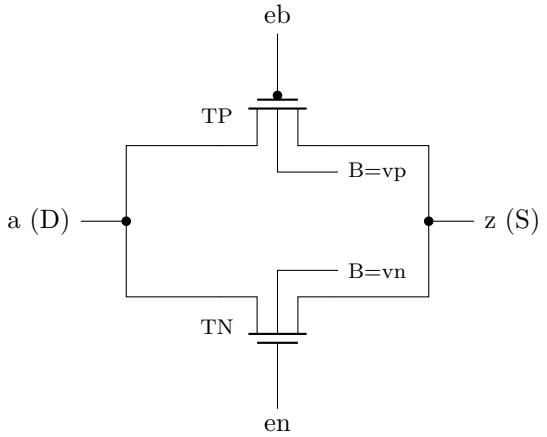


Figure 4: Exact two-MOS transmission gate with all four terminals identified.

The inv cell has $W_P = 3.6k\,\mu$m, $W_N = 1.8k\,\mu$m and $L = 0.25\,\mu$m. The tg cell uses the same widths/length. For tg, both device drains are a and both sources are z; the pass path is bidirectional. TP/TN correspond to MP/MN inside the tg subcircuit. PMOS bodies remain vp and NMOS bodies vn, independent of the signal terminals. The n-channel gate is en, and the p-channel gate is eb. Equal node labels connect the two pass devices in parallel.

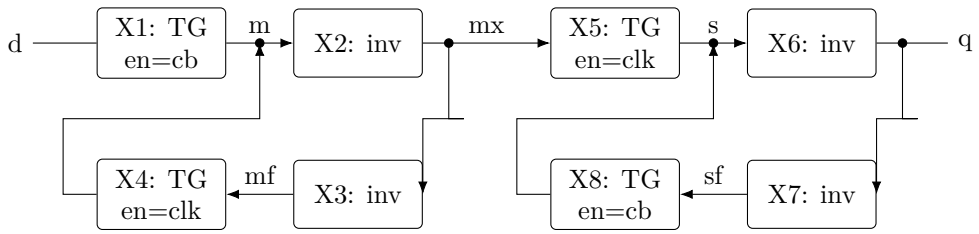


Figure 5: Exact master/slave storage connections. Lower feedback inverters carry signal from right to left.

Every block uses vp/vn rails; CQ=10 fF connects q to vn. cb is the ideal complement of clk. For each TG, eb is the complement of its shown en: X1/X8 have (en,eb)=(cb,clk), X4/X5 have (clk,cb). X1/X4 form the master input/feedback switches and X5/X8 the slave switches. The feedback path is mx $\rightarrow$ X3 $\rightarrow$ mf $\rightarrow$ X4 $\rightarrow$ m; the slave path is q $\rightarrow$ X7 $\rightarrow$ sf $\rightarrow$ X8 $\rightarrow$ s. There are four inv and four tg cells, hence 16 MOS. The separate inverter-chain experiment repeats inv four times with k=1 and a 10 fF load; its edge stimulus is specified in its own netlist.